

Problem

Find all real numbers x satisfying

$$\frac{8^x + 27^x}{12^x + 18^x} = \frac{7}{6}.$$

Source: Original Problem by Aryan Goutam

§1 Extracting the Hidden Pattern

Observe that

$$8^x = (2^x)^3, \quad 27^x = (3^x)^3,$$

and

$$12^x + 18^x = (2^x)^2(3^x) + (2^x)(3^x)^2.$$

Motivated by this structure, let

$$a = 2^x, \quad b = 3^x.$$

Since $a, b > 0$, the given equation becomes

$$\frac{a^3 + b^3}{a^2b + ab^2} = \frac{7}{6}.$$

Multiplying through by $6ab(a + b)$ gives

$$6(a^3 + b^3) = 7ab(a + b).$$

Using the identity

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2),$$

we obtain

$$6(a + b)(a^2 - ab + b^2) = 7ab(a + b).$$

Since $a + b > 0$, we may divide by $a + b$:

$$6(a^2 - ab + b^2) = 7ab.$$

Rearranging,

$$6a^2 - 13ab + 6b^2 = 0.$$

This quadratic factors neatly:

$$(2a - 3b)(3a - 2b) = 0.$$

Therefore,

$$2a = 3b \quad \text{or} \quad 3a = 2b.$$

Substituting back $a = 2^x$ and $b = 3^x$ yields

$$2^{x+1} = 3^{x+1}$$

or

$$2^{x-1} = 3^{x-1}.$$

Since $2^t = 3^t$ holds only when $t = 0$, we obtain

$$x + 1 = 0 \quad \text{or} \quad x - 1 = 0.$$

Hence

$$x = -1 \quad \text{or} \quad x = 1.$$

§2 Verification

For $x = 1$,

$$\frac{8 + 27}{12 + 18} = \frac{35}{30} = \frac{7}{6}.$$

For $x = -1$,

$$\frac{\frac{1}{8} + \frac{1}{27}}{\frac{1}{12} + \frac{1}{18}} = \frac{\frac{35}{216}}{\frac{5}{36}} = \frac{35}{30} = \frac{7}{6}.$$

Both values satisfy the equation.

§3 The Final Answer

Therefore, the set of all real solutions is

$$\boxed{\{-1, 1\}}.$$